

- Processors
  - Micro Processor
  - Micro Controller
- CPU
  - Intel/ AMD, ARM, RISC V
- Memory
  - Volatile (RAM)
  - Non-Volatile (HDD, Flash)
- IO aka Peripherals
  - Keyboard, Mouse
  - Communication Interfaces (WiFi, BT, Ethernet etc)
- Program and Process
- Sections
  - .text .code
  - .data
  - .bss
  - heap
  - stack
- Address Space
  - Kernel Address Space
  - User Address Space
- Booting
  - BIOS
  - Bootloader
  - OS
  - Applications
- Context Switch

- Context Saving + Context Restoring
- Context ?
  - GPR, Flags, PC, SP
- Stack
  - Stack Frame
  - Return Address + Args + local variables
- Privileged
- Non-Privileged
- Header Files and Libraries
- Build Flow
- Toolchain - CPP, Compiler, Linker, objdump, Debuggers
- System Calls
  - Open, Read, Write, Close
- Library Calls
  - fopen, fread, fwrite, fclose
- man
  - man 2 write
  - man -k open
- Libraries
  - Static Library (.a)
  - Dynamic Library
    - Shared Object (.so)
    - Dynamically Linked Library (.dll)
- ar (static)

- ar -crv libname.a file1.o file2.o
- Dynamic (shared)
  - gcc -o libname.so -shared -fPIC file1.o file2.o
- Linking
  - Static Linking
  - Dynamic Linking
    - gcc -o app app.o -L. -lname
    - export LD\_LIBRARY\_PATH=\$LD\_LIBRARY\_PATH:.
    - ./app
- Process States
  - New (Created)
  - Ready
  - Running
  - Waiting
  - Terminated
- Scheduling Algorithms
  - FCFS
  - SJF
  - Round Robin
  - Priority Based
    - Preemptive
    - Non-Preemptive